Topic 17: ELECTROSTATICS
Sub-topic 17.1: Production of Charge
Scenario 1
A printing press in Kampala specializes in high-quality magazine production but faces persistent problems with paper jams and ink smudging during the dry season. The technicians trace the issue to static electricity, as paper sheets moving rapidly through rollers acquire significant electrostatic charges. This causes them to cling to machinery or to each other, disrupting the automated process. The company consults a physics expert who explains the triboelectric effect and recommends installing ionizing bars and maintaining higher humidity in the press room. Implementing these changes based on electrostatic principles has reduced downtime by 30% and improved print quality, demonstrating the importance of understanding charge production in industrial applications.
Task:
As a learner of physics;
(a)Explain the triboelectric effect and how it causes paper to become charged in a printing press.
(b)Describe two methods by which static charge can be produced.
(c)If a polythene rod becomes negatively charged after being rubbed with wool, what charge does the wool acquire? Explain using the law of charge conservation.
(d)Suggest two practical ways to dissipate unwanted static charge in an industrial setting.
Scenario 2
A petrochemical storage facility in the Lake Albert region has implemented strict electrostatic safety protocols after a near-miss incident during fuel transfer. Engineers explain that as hydrocarbon fuels flow through pipes, they can generate static electricity through friction, creating a spark hazard that could ignite flammable vapors. All personnel are now trained in proper grounding and bonding procedures

MR. HILLARY

before any transfer operation. The facility has also installed static dissipaters in storage tanks and requires special footwear and clothing for workers. This comprehensive approach has significantly enhanced operational safety at the facility, protecting both workers and Uganda's valuable petroleum resources.

Task:

As a learner of physics;

(a)Explain how flowing liquids can generate static electricity.

(b)Describe the principle of electrostatic induction.

(c)Why is it dangerous to pump flammable liquids without proper grounding?

(d)A worker wears shoes with insulating soles while handling fuel containers. What risk does this create and how can it be mitigated?

Scenario 3

A physics teacher at a secondary school in Gulu uses simple experiments to demonstrate electrostatic principles. Students rub various materials like glass rods with silk and plastic rods with fur, then use a gold-leaf electroscope to detect and characterize the charges produced. Through these activities, students learn about charge conservation, conduction, and induction. The teacher emphasizes real-world applications, from photocopiers to electrostatic precipitators in industrial smoke stacks, showing how foundational electrostatic concepts underpin important technologies used in Uganda's development.

Task:

As a learner of physics;

(a)Describe how a gold-leaf electroscope works to detect charge.

(b)Explain the difference between charging by conduction and charging by induction.

(c)A charged rod is brought near the cap of an uncharged electroscope and the leaves diverge. What can you conclude about the charge on the rod?

(d)Design a simple experiment to show that there are two types of electric charge.

Scenario 4

A textile factory in Jinja experiences problems with cloth fibers repelling each other during weaving, causing production defects. The issue is particularly severe during Uganda's dry seasons when humidity is low. Factory engineers install static elimination equipment that uses radioactive sources or electrical ionizers to neutralize charges on synthetic fabrics. They also implement humidity control measures in the production areas. Understanding electrostatic principles has enabled the factory to maintain consistent product quality throughout the year, supporting Uganda's growing textile industry and its export markets.

Task:

As a learner of physics;

(a)Explain why synthetic materials are particularly prone to developing static charges.

(b)Describe how humidity affects static electricity accumulation.

(c)Why do like charges repel and unlike charges attract?

(d)Suggest two methods to reduce static electricity problems in a textile factory.

Scenario 5

A meteorological research station on Mount Elgon studies atmospheric electricity, particularly during thunderstorms common in the region. Scientists use specialized equipment to measure electrostatic fields in the atmosphere and correlate them with weather patterns. Their research helps improve lightning prediction models, which is valuable for aviation safety and for protecting isolated communities in mountainous areas. The station also educates local communities about lightning safety, using principles of electrostatics to explain why certain locations and activities are particularly dangerous during thunderstorms.

Task:

As a learner of physics;

MR. HILLARY

(a)Explain how thunderclouds become electrically charged.
(b)Describe the process by which lightning occurs between a cloud and the ground.
(c)Why are tall objects and open areas particularly vulnerable to lightning strikes?
(d)What safety precautions should people take during a thunderstorm based on electrostatic principles?

Sub-topic 17.2: Electrostatic Force

Scenario 6

A research laboratory at Makerere University is using an updated version of Coulomb's torsion balance experiment to precisely measure electrostatic forces. Physics students set up the apparatus with small charged spheres and measure the force between them at varying distances. Their careful measurements verify the inverse-square law relationship that Coulomb first established. This hands-on experience helps students appreciate the foundational nature of Coulomb's work, which underpins much of modern electrical science and technology. The laboratory exercise also highlights the challenges of making precise physical measurements and the importance of controlling variables in experimental physics.

Task:

As a learner of physics;
(a)State Coulomb's Law in both words and mathematical form.
(b)Two point charges of +2 μC and +3 μC are placed 0.1 m apart in air. Calculate the electrostatic force between them.
(c)If the distance between the two charges in part (b) is doubled, what happens to the force between them?
(d)Explain why Coulomb's Law is described as an inverse-square law.

Scenario 7

An industrial process at a cement factory in Tororo uses electrostatic forces to remove dust particles from exhaust gases. Engineers design plates with high voltage differences that create strong electric fields, causing dust particles to become charged and migrate toward collection plates. This electrostatic precipitation technology helps the factory meet Uganda's environmental standards by significantly reducing particulate emissions. The plant technicians must regularly maintain the system and understand the relationship between voltage, plate separation, and the resulting electrostatic forces to ensure optimal performance of the pollution control equipment.

Task:

As a learner of physics;
(a)Explain how an electrostatic precipitator uses Coulomb forces to remove dust from air.
(b)Two parallel plates have a potential difference of 10 kV and are separated by 5 cm. Calculate the electric field strength between them.
(c)How does the electrostatic force on a charged particle depend on the electric field strength?
(d)Why is electrostatic precipitation an effective method for controlling industrial air pollution?

Scenario 8

A technology company in Kampala is developing an electrostatic sorting system for separating different types of plastic waste for recycling. The system uses the fact that different plastics acquire different charges when rubbed, allowing them to be separated by electrostatic forces. This technology could significantly improve Uganda's waste management capabilities, particularly for dealing with the growing problem of plastic pollution. Understanding the relationship between material properties and electrostatic behavior is crucial for optimizing the sorting process and making plastic recycling more economically viable in the Ugandan context.

Task:

As a learner of physics;

MR. HILARY

(a)Explain how electrostatic forces can be used to separate different materials.
(b)Two identical spheres with charges +5 μC and -3 μC are brought into contact and then separated. What is the final charge on each sphere?
(c)Calculate the force between two charges of +4 μC and -6 μC placed 15 cm apart in air.
(d)Why is electrostatic separation particularly useful for recycling certain materials?

Scenario 9

A physics education initiative uses simple experiments to help students visualize electrostatic forces. Students charge small suspended pith balls and observe how the distance between them affects the repulsive force. They also explore how introducing different materials between charged objects affects the force. These experiments make abstract concepts tangible and help students understand the fundamental nature of electrostatic interactions. The program has been particularly successful in rural schools, where teachers have developed low-cost apparatus from locally available materials to demonstrate these important physical principles.

Task:

As a learner of physics;
(a)Describe an experiment to demonstrate that electrostatic force depends on the distance between charges.
(b)How does the medium between charges affect the electrostatic force between them?
(c)Two charges experience a force F when placed a distance d apart in air. What would be the force if they were placed the same distance apart in a medium with dielectric constant 4?
(d)Why is it important for students to perform experiments with electrostatic forces?

Scenario 10

An inkjet printer manufacturer establishing a plant in the Namanve Industrial Park must carefully control electrostatic forces to ensure precise droplet placement. The printing technology uses electrostatic fields to deflect charged ink droplets onto specific positions on paper. Engineers must calculate the exact field strengths needed for different printing resolutions and ink properties. This application of Coulomb's Law is essential for producing high-quality printed materials, supporting Uganda's growing publishing and packaging industries. The technology represents a sophisticated industrial application of fundamental electrostatic principles.

Task:

As a learner of physics;
(a)Explain how electrostatic forces are used in inkjet printing technology.
(b)A charge of 1.6×10^{-13} C experiences a force of 8×10^{-9} N in an electric field. Calculate the electric field strength.
(c)How would the deflection of an ink droplet change if its charge were doubled while the electric field remained constant?
(d)Why is precise control of electrostatic forces important in modern printing technology?

Topic 18: CAPACITORS

Sub-topic 18.1: Parallel Plate Capacitors

Scenario 11

A radio repair workshop in Kampala frequently deals with faulty tuning capacitors in older radio models. The technicians need to understand how parallel plate capacitors work to diagnose and fix issues with station selection. They explain to apprentices that by changing the overlap area of the plates, the capacitance changes, allowing the radio to select different frequencies. The workshop has become a training ground for young electronics enthusiasts, many of whom go on to pursue careers in Uganda's

MR. HILARY

growing electronics industry. The practical understanding of capacitors has helped preserve older radio technology while teaching fundamental principles that apply to modern communication devices.

Task:

As a learner of physics;

(a)Define capacitance and state its SI units.

(b)Calculate the capacitance of a parallel plate capacitor with plates of area 0.2 m^2 separated by 0.5 mm of air.

(c)Explain how the capacitance changes if the distance between plates is doubled.

(d)Why are variable capacitors essential in radio tuning circuits?

Scenario 12

A physics teacher at a school in Jinja uses homemade parallel plate capacitors to demonstrate energy storage principles. Students construct simple capacitors using aluminum foil sheets separated by paper dielectric materials. They measure how different dielectrics affect capacitance using multimeters and simple circuits. This hands-on approach helps students grasp abstract concepts like permittivity and energy storage. The activity has sparked interest in practical physics, with several students developing projects for the annual Uganda Science and Innovation Competition based on capacitor applications.

Task:

As a learner of physics;

(a)Explain the function of a dielectric material in a capacitor.

(b)A parallel plate capacitor has a capacitance of 10 μF in air. Calculate its capacitance when filled with a dielectric of constant 5.

(c)Describe how the stored energy changes when a dielectric is inserted into a charged capacitor.

(d)Design an experiment to compare the dielectric constants of different materials.

Scenario 13

A solar energy company in Mbale uses large capacitor banks to smooth out power fluctuations in their solar installations. Engineers must calculate the required capacitance to maintain stable voltage outputs despite varying sunlight conditions. The capacitors store excess energy during peak production and release it during brief cloud cover, improving the reliability of solar power systems in rural health centers and schools. This application has made solar energy more practical for critical applications in Uganda's off-grid communities.

Task:

As a learner of physics;

(a)Derive the formula for the capacitance of a parallel plate capacitor.

(b)Calculate the area needed for a parallel plate capacitor with air dielectric to have a capacitance of 1μF if the plates are 1 mm apart.

(c)Explain how capacitors help stabilize power supply in renewable energy systems.

(d)Why are capacitor banks important in modern energy systems?

Scenario 14

A medical equipment maintenance team at a regional hospital in Gulu troubleshoots defibrillator machines that use capacitor banks to store the energy needed for heart resuscitation. Technicians must understand capacitor charging and discharging principles to ensure these life-saving devices function reliably. They perform regular tests to verify that the capacitors can store and deliver the precise energy levels required for effective defibrillation. This knowledge has helped maintain critical medical equipment in regions with limited technical support resources.

Task:

As a learner of physics;

(a)Explain how capacitors store energy in a defibrillator.

MR. HILARY

(b)A defibrillator capacitor is charged to 5000 V and has a capacitance of 100 μF. Calculate the energy stored.
(c)Describe what happens to the stored energy when the capacitor is discharged through a patient's chest.
(d)Why is precise energy control important in medical devices like defibrillators?

Scenario 15

An automotive electronics workshop in Kampala specializes in repairing car audio systems where capacitors are used extensively in crossover networks and power supply filters. Technicians measure capacitor values and test their performance in different circuit configurations. Understanding how parallel plate capacitors work helps them diagnose issues with sound quality and power delivery in vehicle entertainment systems. This expertise supports Uganda's growing automotive services industry, particularly as more sophisticated electronic systems become common in imported vehicles.

Task:

As a learner of physics;
(a)Explain how capacitors are used in audio system crossover networks.
(b)Calculate the charge stored on a 1000 μF capacitor when connected to a 12 V car battery.
(c)Describe what happens to a capacitor's ability to store charge if the plate area is increased.
(d)Why are capacitors important in automotive electronics?

Sub-topic 18.2: Charging and Discharging Capacitors

Scenario 16

A photography studio in Entebbe uses professional flash units that rely on capacitor charging and discharging principles. The photographers need to understand the timing involved in charging the capacitors to ensure ready availability for rapid-sequence shooting. They've learned that the RC time constant determines how quickly the flash units recycle between shots. This understanding has improved their efficiency during important events like weddings and corporate functions, enhancing the quality of professional photography services in Uganda's growing creative industry.

Task:

As a learner of physics;
(a)Define the time constant in an RC circuit.
(b)A 100 μF capacitor is charged through a 10 kΩ resistor. Calculate the time constant.
(c)Explain what percentage of the full charge the capacitor reaches after one time constant.
(d)Why is understanding charging times important in photographic flash equipment?

Scenario 17

A university electronics laboratory in Mbarara studies capacitor discharging through various resistive loads. Students use oscilloscopes to visualize the exponential decay of voltage during discharge and verify the theoretical time constants. These experiments help bridge the gap between abstract mathematical models and observable circuit behavior. The skills developed in these labs prepare students for careers in Uganda's emerging electronics manufacturing sector, where timing circuits are essential for many applications.

Task:

As a learner of physics;
(a)Derive the expression for voltage during capacitor discharge.
(b)A capacitor is discharged through a resistor. If the initial voltage is 12 V, calculate the voltage after one time constant.
(c)Describe how the discharge curve would change if the resistance value were increased.
(d)Why is exponential decay characteristic of capacitor discharge?

MR. HILARY

Scenario 18
A traffic light control system in Kampala uses timing circuits based on capacitor charging to manage light sequences at intersections. Engineers must calculate precise RC values to ensure appropriate timing for different traffic conditions. The system's reliability depends on understanding how capacitors charge through resistors to trigger switching circuits at predetermined intervals. This application of capacitor charging principles has contributed to improved traffic flow and road safety in Uganda's capital city.
Task:
As a learner of physics;
(a)Explain how RC circuits can be used as timing devices.
(b)A timing circuit needs a delay of 5 seconds. If using a 100 μF capacitor, calculate the required resistance.
(c)Describe what happens in an RC circuit when the capacitor is fully charged.
(d)Why are timing circuits important in traffic control systems?
Scenario 19
A cardiac care unit at a hospital in Jinja uses heart rate monitors that employ capacitor charging circuits to measure pulse intervals. Medical technicians need to understand how these circuits work to properly calibrate and maintain the equipment. The capacitors charge at rates proportional to heart rate, providing visual and audible signals for medical staff. This application of capacitor charging principles supports critical healthcare services in Uganda's regional hospitals.
Task:
As a learner of physics;
(a)Explain how capacitor charging can be used to measure time intervals.
(b)Calculate how long it takes for a capacitor to charge to 95% of the supply voltage in terms of the time constant.
(c)Describe the factors that affect the charging rate of a capacitor.
(d)Why are reliable timing circuits important in medical monitoring equipment?
Scenario 20
A renewable energy research center in Uganda studies capacitor charging and discharging in the context of energy storage for small-scale solar systems. Researchers investigate how capacitors can provide brief bursts of power for starting motors or handling short-term load spikes in off-grid power systems. This research explores alternatives to batteries for certain applications, particularly in remote areas where battery maintenance and replacement present challenges.
Task:
As a learner of physics;
(a)Compare capacitor energy storage with battery energy storage.
(b)A capacitor is charged to 50 V and then discharged through a 100 Ω resistor. If the capacitance is 500 μF, calculate the initial discharge current.
(c)Explain why capacitors can deliver high power for short durations.
(d)How can capacitor energy storage complement battery systems in renewable energy applications?
Sub-topic 18.3: Energy Storage in Capacitors
Scenario 21
A physics demonstration at a Kampala science center uses a large capacitor bank to show dramatic energy discharge through a flash tube. Educators explain how the capacitors store electrical energy in an electric field and release it suddenly to create a brilliant flash of light. The demonstration helps visitors understand energy storage principles and their applications in technologies like camera flashes and strobe

MR. HILARY

lights. This public engagement activity has increased interest in physics among secondary school students visiting the center.

Task:

As a learner of physics;

(a)Derive the formula for energy stored in a capacitor.

(b)Calculate the energy stored in a 1000 μF capacitor charged to 400 V.

(c)Explain where the energy is physically stored in a capacitor.

(d)Why are capacitors able to release energy much faster than batteries?

Scenario 22

An industrial plant in Tororo uses capacitor banks to improve power factor in their electrical system, reducing energy costs. Engineers explain that the capacitors store and release energy to counteract the effects of inductive loads like electric motors. This application of capacitor energy storage principles has significantly reduced the plant's electricity bills while stabilizing the power quality. The success of this project has led other Ugandan industries to adopt similar power factor correction measures.

Task:

As a learner of physics;

(a)Explain how capacitors can improve power factor in AC systems.

(b)A capacitor stores 0.5 J of energy at 200 V. Calculate its capacitance.

(c)Describe why industrial facilities use capacitor banks for power factor correction.

(d)How does power factor correction benefit both the consumer and the power utility?

Scenario 23

A research project at Makerere University investigates using capacitors for energy storage in small-scale renewable systems. Students compare different capacitor technologies - electrolytic, ceramic, and super capacitors - for their energy density and discharge characteristics. The research aims to develop affordable energy storage solutions for rural applications where conventional batteries may be unsuitable. This work supports Uganda's goals of increasing renewable energy adoption, particularly in off-grid communities.

Task:

As a learner of physics;

(a)Explain the difference between conventional capacitors and super capacitors.

(b)A supe capacitor has a capacitance of 100 F and is charged to 2.5 V. Calculate the stored energy.

(c)Compare the energy density of capacitors with different dielectric materials.

(d)Why is research into capacitor energy storage important for renewable energy development?

Scenario 24

An electronics manufacturer in the Namanve Industrial Park produces power supply units that use capacitors to smooth DC output voltage. Quality control technicians test the capacitors' ability to store and release energy to maintain stable voltages under varying load conditions. Understanding energy storage principles helps them select appropriate capacitors for different applications, ensuring the reliability of electronic equipment produced for the Ugandan market and for export.

Task:

As a learner of physics;

(a)Explain how capacitors smooth DC power supply outputs.

(b)A power supply requires a capacitor to store 0.1 J of energy and operate at 25 V. Calculate the minimum capacitance needed.

(c)Describe what happens to the output voltage if the smoothing capacitor fails.

(d)Why are capacitors essential components in power supply circuits?

MR. HILARY

Scenario 25
A vocational training institute in Lira teaches automotive electronics, including the role of capacitors in vehicle ignition systems. Students learn how capacitors store energy that is suddenly discharged to create sparks in spark plugs. This practical understanding helps them diagnose and repair ignition problems in the many imported used vehicles on Uganda's roads. The skills developed through this training support Uganda's automotive repair industry and provide employment opportunities for technical graduates.
Task:
As a learner of physics;
(a)Explain how capacitors are used in automotive ignition systems.
(b)An ignition capacitor stores 0.05 J of energy at 300 V. Calculate its capacitance.
(c)Describe why rapid energy discharge is important in ignition systems.
(d)How does understanding capacitor energy storage help automotive technicians?

Topic 19: DIGITAL ELECTRONICS
Sub-topic 19.1: Junction Diodes
Scenario 26
A solar power installation company in Uganda uses junction diodes in their photovoltaic systems to prevent reverse current flow at night. Engineers explain to installation technicians how diodes act as one-way valves for electric current, ensuring that solar panels don't drain batteries when not generating power. This understanding has improved the reliability of solar installations across Uganda, particularly in rural areas where maintenance visits are infrequent. The company has developed training materials that use simple analogies to help technicians with varying educational backgrounds understand and apply diode principles in their work.
Task:
As a learner of physics;
(a)Explain the working principle of a p-n junction diode.
(b)Draw the circuit symbol for a diode and indicate the direction of conventional current flow.
(c)Describe how diodes are used in solar power systems.
(d)Why is reverse current protection important in solar energy systems?
Scenario 27
An electronics repair workshop in Kampala's downtown area specializes in fixing power supplies and battery charters where diodes are commonly used in rectifier circuits. The technicians use multimeters to test diodes for proper operation, checking forward voltage drop and reverse leakage current. Their ability to diagnose diode-related issues has made the workshop a go-to resource for repairing electronic equipment in the community. This practical expertise supports the circular economy by extending the life of electronic devices in Uganda.
Task:
As a learner of physics;
(a)Describe how to test a diode using a multimeter.
(b)Explain what is meant by the "forward voltage drop" of a diode.
(c)A silicon diode has a forward voltage drop of 0.7 V. If it conducts 100 mA, calculate the power dissipated.
(d)Why are diodes essential components in power supply rectifier circuits
Scenario 28
A physics teacher at a school in Fort Portal uses simple experiments with diodes, batteries, and LEDs to demonstrate semiconductor principles. Students build basic circuits that show how diodes only conduct in one direction and learn to identify the anode and cathode terminals. These hands-on activities make

MR. HILLARY

abstract semiconductor concepts tangible and help students understand the foundation of modern electronics. The school has incorporated these experiments into the physics curriculum, improving student engagement with the subject.

Task:

As a learner of physics;

(a)Draw and explain the I-V characteristic curve of a junction diode.

(b)Describe the difference between forward bias and reverse bias in a diode.

(c)Design a simple circuit to demonstrate the rectifying action of a diode.

(d)Why are hands-on experiments important for understanding electronic components?

Scenario 29

A telecommunications company in Uganda uses diodes in various protection circuits to safeguard sensitive equipment from voltage spikes and static electricity. Engineers design these circuits using Zener diodes for voltage regulation and other diodes for preventing reverse polarity connection. Understanding diode characteristics helps them create robust protection systems that maintain service reliability, particularly during Uganda's frequent thunderstorms that can induce power surges in communication infrastructure.

Task:

As a learner of physics;

(a)Explain how a Zener diode differs from a regular diode.

(b)Describe how diodes can protect electronic circuits from voltage spikes.

(c)A Zener diode is rated at 5.1 V. What is its function in a voltage regulation circuit?

(d)Why is circuit protection important in telecommunications equipment?

Scenario 30

A renewable energy project in rural Uganda uses diodes in simple charge controllers for solar home systems. Community technicians are trained to understand how diodes prevent battery over discharge and manage power flow between solar panels, batteries, and loads. This knowledge empowers local communities to maintain their own energy systems, supporting sustainable development in off-grid areas. The project has successfully deployed hundreds of solar systems with local technicians providing ongoing support.

Task:

As a learner of physics;

(a)Explain how diodes are used in solar charge controllers.

(b)Describe what happens to a diode when it is reverse biased beyond its breakdown voltage.

(c)Why is it important for community technicians to understand basic electronic components?

(d)How can simple diode-based circuits support renewable energy adoption in rural areas?

Sub-topic 19.2: Transistors

Scenario 31

A radio station in Mbale uses transistor amplifiers in their broadcast equipment to boost audio signals before transmission. The technical team understands how transistors provide signal amplification through their ability to control large currents with small input signals. This knowledge helps them maintain clear broadcast quality despite varying signal conditions. The station has become a training ground for aspiring broadcast engineers in eastern Uganda, with many technicians learning transistor fundamentals through hands-on experience with the transmission equipment.

Task:

As a learner of physics;

(a)Explain the basic operation of a bipolar junction transistor as an amplifier.

MR. HILARY

(b)Draw the circuit symbols for NPN and PNP transistors.
(c)Describe how a transistor can control a large current with a small input current.
(d)Why are transistors essential in audio amplification systems?

Scenario 32

An agricultural research center in Kabale uses transistor-based temperature control systems in their seed storage facilities. Engineers design circuits that use transistors as switches to activate cooling systems when temperatures rise above predetermined levels. Understanding transistor switching characteristics helps them create reliable environmental control systems that preserve valuable seed varieties for Uganda's farming communities. This application of transistor technology supports food security by maintaining genetic diversity in crop species.

Task:

As a learner of physics;
(a)Explain how a transistor operates as a switch.
(b)Describe the cutoff and saturation regions of transistor operation.
(c)Design a simple transistor switch circuit to control a relay based on temperature input.
(d)Why are transistor switches preferred over mechanical switches in control systems?

Scenario 33

A vocational training institute in Kampala teaches transistor principles as part of their electronics technician program. Students learn to identify transistor terminals, test transistors using multimeters, and build simple amplifier and switch circuits. The practical skills developed in these courses prepare students for employment in Uganda's growing electronics sector, from equipment repair to industrial automation. Many graduates have found employment with telecommunications companies, broadcast stations, and manufacturing plants across Uganda.

Task:

As a learner of physics;
(a)Describe how to identify the base, collector, and emitter of a bipolar junction transistor.
(b)Explain the concept of current gain (β) in a transistor.
(c)A transistor has a base current of 20 μA and a collector current of 2 mA. Calculate the current gain.
(d)Why are practical skills in transistor circuits valuable for employment in the electronics industry.

Scenario 34

A solar lighting company in Uganda uses transistors in the control circuits of their advanced solar lanterns and home lighting systems. Engineers design circuits that use transistors for maximum power point tracking, battery charging control, and LED dimming functions. Understanding transistor characteristics helps them optimize the efficiency and functionality of their products, making solar lighting more affordable and effective for Ugandan households. The company's success has contributed to increased adoption of solar lighting in areas with limited grid electricity.

Task:

As a learner of physics;
(a)Explain how transistors can be used for power control in electronic devices.
(b)Describe the advantages of using transistors instead of mechanical switches in power control applications.
(c)Why are transistors important in maximizing the efficiency of solar-powered devices?
(d)How has transistor technology contributed to the adoption of renewable energy solutions

Scenario 35

A university engineering department in Mbarara studies transistor applications in simple computing circuits. Students build basic logic gates using transistors and learn how this form the foundation of digital computers. This fundamental understanding prepares students for advanced studies in computer

MR. HILLARY

engineering and supports Uganda's goals of developing local expertise in information technology. The department has established partnerships with local tech companies to ensure the curriculum remains relevant to industry needs.

Task:

As a learner of physics;

(a)Explain how transistors can be used to create basic logic gates.

(b)Design a simple AND gate using transistors and diodes.

(c)Describe the significance of transistors in the development of modern computing.

(d)Why is understanding transistor fundamentals important for computer engineering?

GOOD LUCK, MR. HILLARY

MR. HILLARY